

Digital Logic Design (EL-1005)

Spring 2024

Project Description & Evaluation Criteria

Intro

The DLD project based on combinational/Sequential logic is to be assessed as lab project. Project is to be done in a group of two students. Project is to be implemented on both Proteus and breadboard. A separate DC power supply like battery of 9V should be required for hardware demo. Students need to purchase their own hardware based on the project requirements. After the completion of the project you need to write a report that summarizes the whole project, along with the screenshots of the Proteus simulations and then hardware implementation.

List of Projects

You need to choose just one project from the following list:

1. Traffic light system:
Traffic light system for a junction of 4 roads, 2 main roads and 2 side roads. The green signal goes for a longer time for the main roads and shorter for the side roads.
2. Digital Clock:
A digital clock that displays seconds and minutes, starting from time 00:00, increases seconds as a clock works and then after 60 seconds increases the minutes. You may decrease the time between 2 seconds in order to save time during demo.
3. Digital Counter:
A digital counter that counts the number of times a push button is pressed from 0 till 99. There must be a reset button that resets the counter anytime.
4. Stop watch:
A stop watch that measures the time as a typical stop watch works.
5. Voting System:
A voting system that shows the win or lose based on the number of votes out of 5. There will be 5 push buttons to vote. If the number of votes are 3 out of 5 then a green LED turns on that shows the win and if there are 2 votes out of 5 then a red LED turns on that shows the lose.

Project Grouping and Proposal

You need to make a group of 2 students and then choose the project you'll be working on from the above list. Make a document that clearly mentions the names and roll numbers of the group members along with the title of the project you choose. Submit the document in GCR in Proposal Submission tab. Please note that only one member of the group needs to submit the proposal document. Make sure the groups you make are not frequently changed.

Work Distribution and Collaboration

Both group partners are expected to work on the project together, communicate effectively and share the workload in a fair manner. There will be a viva or other means to determine the veracity of any claim. Free loaders will be penalized.

Project Phases & Assessment

	Marks	
Overall project Weightage	20 absolutes	
Within project		
Proteus + Breadboard implementation + Report	5 + 17 + 3	
Project Phases for submission		Deadlines
Phase 1 – Proteus Simulation + demo	3 + 2	
Phase 2 – Hardware Working + Efficient Wiring + viva/demo + Report	10 + 2 + 5 + 3	May 13th
Total	20	

Hardware must be submitted physically and report must be submitted to classroom 13th May.

Demo Dates will be announced later.

Proteus Build & Simulation

Start with Proteus simulation of your project, resolving all bugs and interfacing issues. You must submit your code and Proteus files (if any) to the lab instructor for plagiarism check and demo run.

Project Viva

Each group partner must be prepared to answer all aspects of the project even if they did not do that part themselves. – be prepared.

Report of Project

Each group must need to submit a complete report till 13th May, 2024. Report will include following items:

- Title Page including Project Name, student's identity and logo of your project.
- Must include an outline in the report where page indexing will be used to go to specific page number to look at certain features of the project.
- It must include complete description/details of your project supported with circuit diagrams, block diagrams, flow charts, Proteus diagrams, hardware components required and breadboard images loaded with components on it.
- It must also include images of completed project showing clearly that your project is working fine. Every aspect of the project should be included as images in the document with a separate heading like Simulation Results etc.
- Report must include the procedure which you have followed to assemble the circuit on Proteus and on breadboard as well.
- It must include separate page for Bibliography/references where all references should be there from where you gather information regarding project.
- Last but not least, every minor aspect which you think is important during project must also include in the report.

Regarding Project Submission

- Each group must follow the deadline of the classroom. Late submission will result in complete deduction of marks of specific phase and that submission will not be graded at all.
- At the last deadline, a video recordings of hardware running must also be uploaded on GCR
- Just one group member needs to submit all the phase submissions.